

Exploring the Impact of Drivers' Emotion and Multi-task Learning on Takeover Behavior Prediction in Multimodal Environment

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Abstract

With the development of autonomous driving technology, drivers may be immersed in non-driving related tasks. Therefore, takeover safety has received increasing attention. Previous studies have focused on designing the empirical analysis to formulate preventive measures or constructing methods for a single-task learning strategy. However, due to the persistent gaps between modalities and ignoring the correlation between tasks, they make it difficult to capture similar information between modalities. In this paper, we propose the Multi-TBP, a general deep learning framework for takeover behavior prediction. Experiments show that our method achieves the best performance on all tasks. Through ablation studies and visualization, we find that each modality's importance varies from each task, and there are also correlations between tasks.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Automated driving, Takeover behavior, Emotion, Multimodal fusion, Multi-task learning

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1 Introduction

With the development of autonomous driving technology, drivers can gradually engage in various non-driving related tasks (NDRTs) to enhance the driving experience, such as watching movies, chatting or reading [25, 31]. However, due to legal or technical restrictions, when faced with specific traffic situations, the driver needs to takeover the vehicle immediately, so the vehicle will issue the takeover requests (TOR) and warn in the form of vision, hearing or touch [1, 27, 45]. When the vehicle needs to takeover urgently, due to the driver's participation in various NDRTs, the current attention cannot meet the requirements (for example, it is difficult to perceive TOR when chatting), resulting in poor takeover quality and posing a hidden danger to driving safety [51, 52]. Therefore, the key challenge is how to transfer control from the vehicle to the driver smoothly [18].

For the above challenge, many previous studies have focused on the empirical analysis of driver takeover time and quality, involving exploring how the external driving environment [9, 16], types of NDRTs [41], driver features [4, 35], and human-machine interaction interface [21] affect takeover behavior, and formulating preventive measures based on this. However, since the above methods are difficult to help vehicles predict whether the driver can quickly handle takeover events, it poses a hidden danger to driving safety. Therefore, some studies have gradually begun to use simple machine learning (ML) methods [10, 12, 24] and deep learning (DL) methods [7, 11, 18, 31] to help vehicles predict their takeover behavior (such as takeover readiness [2], reaction time [34] and quality [18]) based on the driver's physical state data. These methods aim to enable the vehicle to adjust TOR by predicting the driver's takeover behavior, thereby smoothly transferring control and improving takeover safety. However, due to the complexity of multimodal data from drivers [20] and the challenge of persistent gaps between modalities [15, 48], they are difficult to capture similar information between modalities when performing multimodal fusion, making modality interaction difficult. Meanwhile, existing methods usually use a single-task learning strategy, while the information expressed by the same data when performing different tasks often shows a

certain correlation [17, 50], so this will further limit the performance of the models. In addition, previous studies have examined the impact of drivers' emotions on takeover behavior and found that it usually plays an important role [11, 13, 18]. However, the interaction and importance differences between emotion and other multimodal data have not been fully explored, making it difficult for models to model them.

To address the above issues, we propose Multi-TBP, a general deep learning framework for takeover prediction that uses fine-grained body motion data of the driver. Multi-TBP adjusts the weights of each modality during fusion through the Cross-modal Transformer to capture the correlation between modalities. Meanwhile, unlike existing studies, to explore the correlation between different evaluation criteria for driver behavior in the same takeover environment, we construct a multi-task learning strategy that can use the same features to simultaneously learn and predict different takeover tasks, including takeover readiness (whether it is ready to accept TOR), takeover reaction time (the time to change to manual driving after TOR), and takeover quality (the driver's ability to handle takeover). We tested Multi-TBP using the latest dataset [18] and achieved state-of-the-art (SOTA) performance on all takeover tasks. Then, we conducted several ablation studies and visualizations and found that (1) for the prediction of takeover behavior, the importance of each modality varies for different tasks; (2) there is also a specific correlation between different takeover tasks. This work aims to provide insights into the distribution of modality importance and task correlation for subsequent research. Our main contributions are as follows:

- We propose Multi-TBP, a general deep learning framework for takeover behavior prediction.
- Design a multi-task learning strategy to learn and predict multiple takeover tasks simultaneously.
- Further explore the importance distribution of driver information including emotions in different tasks, and the correlation between different takeover tasks, to provide suggestions for subsequent research.

2 Related Work

With the development of autonomous driving technology, drivers tend not to participate in driving tasks actively and may be immersed in NDRTs. Therefore, takeover safety has received increasing attention. Current research mainly uses takeover reaction time [28, 32, 43], quality [52] and readiness [2, 7] to measure the safety of takeover behavior.

Previous research mainly focused on empirically analysing drivers' takeover behavior prediction. For instance, Zhang et al. [47] found that takeover time is related to the urgency of the driving situation, NDRT, etc. Feldhütter et al. [14] found that NDRTs may increase reaction time. Furthermore, many studies have also found that various driving environments, driving expectations [33], TOR [36, 38], age [40] and gender [42] will affect takeover reaction time. Then, some studies found that takeover quality may be affected by factors such as driver cognitive burden [9, 46], emotions and trust [8, 11], and NDRTs [26]. Due to the above methods cannot help the vehicle predict whether the driver can quickly handle the takeover

event, resulting in a hidden danger to driving safety. Therefore, predicting takeover behavior through computational models (machine learning and deep learning) has attracted widespread attention recently. For machine learning, Lotz and Weissenberger [24] applied a linear support vector machine (SVM) method to predict takeover time using drivers' eye movement and body posture. Du et al. [10, 12] developed a random forest (RF) model based on drivers' physiological data and environmental parameters to analyze takeover quality. In addition, k-nearest neighbor (KNN) and Naive Bayes were also used to predict takeover time [12]. In deep learning, Deo and Trivedi [7] used a long short-term memory (LSTM) to estimate the drivers' takeover readiness. At the same time, some works [11, 18, 31] also predicted multiple takeover behaviors by constructing integrated frameworks based on deep neural networks.

Among the existing studies that are most relevant to us, Du et al. [9] analyzed the laws of cognitive load and takeover behavior and analyzed the data through random forests [10], but faced the problems of gap between data and ignoring the relevance of modeling tasks, which affected the real-time analysis effect. Du et al. [11] and Gu et al. [18] explored the impact of emotional valence and arousal on takeover behavior and introduced DL models to analyze data, but mainly relied on behavioral data statistics and lacked exploration of the modeling of emotion representation and its importance distribution. In contrast, Multi-TBP bridges the data gap and captures the intrinsic connection of tasks through multimodal fusion modules and multi-task learning strategies, while combining visualization to explore the laws of emotional modeling.

In summary, existing research mainly focuses on reasonably collecting data and using basic models to predict. However, due to the challenge of the persistent gap between multimodal data and the correlation between multiple tasks, the performance of above methods is often limited. Therefore, this paper proposes Multi-TBP, which effectively bridges the gap between modalities through a multimodal fusion based on Transformer, clarifies that driver emotions dominate modal interaction, and provides a unified framework for takeover prediction through a multimodal fusion method with no restriction on the number of modalities and a multi-task learning strategy.

3 Methodology

3.1 Data Acquisition

To explore the importance of driver emotions and correlation of each task, we selected the latest EmoTake dataset [18] that recruited 26 participants and used watch video clips as NDRT to deliver TOR reminders auditorily (external audio speakers) and visually (pop-up windows). The dataset details are described as follows:

Data types. EmoTake leverages a camera to collect four types of information. **Action unit.** MTCNN [49] extracts facial expression features from a camera, and Action Units [37] is chosen to encode the face via OpenFace 2.0. **Eye movement and Head posture.** OpenFace 2.0 obtains each frame's eye movement features and head pose features. The eye movement are the direction of the left and right eye gaze and the angle of both eyes. The head pose includes the head position relative to the camera and the rotation angles. **Body posture.** The OpenPose [30] is used to detect drivers' limbs in frame images and provide 12-point coordinates.

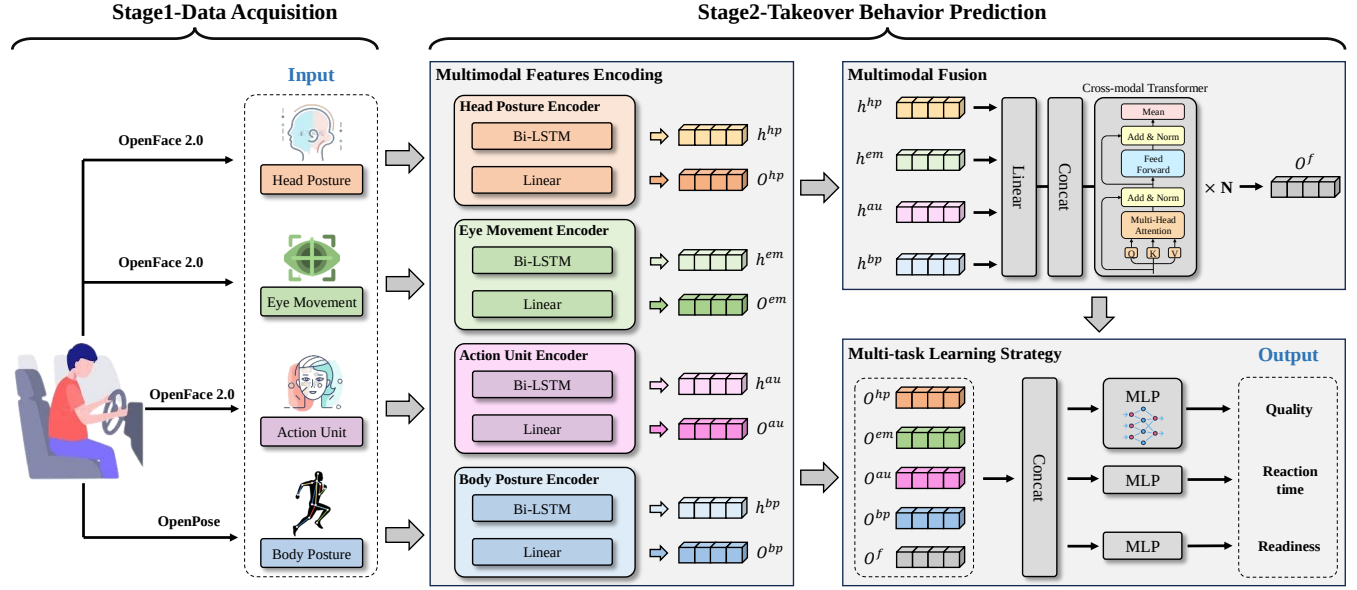


Figure 1: The workflow of Multi-TBP consists of data acquisition and takeover behavior prediction. The orange, green, purple, blue and gray represent the processing of head posture, eye movement, action unit, body posture and multimodal fusion.

Data labels. This dataset has three labeling types. **Takeover quality.** The takeover quality measure is chosen to minimize the individual differences. It has a 5-point scale for representing the different levels of quality rated (‘‘Level 1’’ = worst, ‘‘Level 5’’ = perfect). **Takeover reaction time.** It is defined as the duration between a TOR occurring and the takeover control transition. It is divided into ‘‘Short’’, ‘‘Medium’’, and ‘‘Long’’ when reaction time $\in (0, 1.22s]$, $(1.22s, 2.10s]$ and $(2.10s, 3.5s]$. **Takeover readiness.** It is labeled into binary outcomes (‘‘Ready’’ and ‘‘Not Ready’’).

3.2 Takeover Behavior Prediction

Figure 1 shows the workflow of Multi-TBP. It uses Cross-modal Transformer to perform multimodal fusion and effectively explore the correlation between modalities. Meanwhile, a multi-task learning strategy mines information related to different tasks in unified data. Specifically, since the collected data include time series information, we first use bidirectional LSTM to build a multimodal feature encoder module to learn the contextual features of each frame h^m and the segment o^m , $m \in au, em, hp, bp$. Then, we use Cross-modal Transformer to design a multimodal fusion module to adjust the importance of modalities through fine-grained features to perform multimodal fusion and further bridge the gap between modalities. To avoid LSTM forgetting long-time series information, we first use Linear to compress the contextual information of each frame in length, and then input unimodal features into the Cross-modal Transformer, utilising the interaction of Q, K and V to change the importance of each modality and output unified multimodal features O^f . The same data will express different information when performing different tasks, and this information is often relevant. Thus, we finally construct a multi-task learning strategy that integrates the fragment context information of each

modality and unified multimodal features, and mines the correlation information between tasks by executing and integrating the predictions of multiple tasks simultaneously. The final loss $\mathcal{L}$ is defined as follows:

$$\mathcal{L} = \alpha \mathcal{L}_{qu} + \beta \mathcal{L}_{ra} + \gamma \mathcal{L}_{re} \quad (1)$$

where $\mathcal{L}_{qu}$, $\mathcal{L}_{ra}$ and $\mathcal{L}_{re}$ are the task loss of quality, reaction time and readiness respectively.

In experimental settings, we used 3-layer Cross-modal Transformer, the cross-entropy loss function, and an Adam optimizer with a learning rate 0.0005 to train Multi-TBP. We used the grid search method to find the optimal hyperparameter set for Multi-TBP. The values of α , β and γ are $[1, 0.1, 0.1]$, $[0.2, 1, 0.2]$ and $[0.2, 0.2, 1]$ respectively, in quality, reaction time, and readiness. The epoch is set to 40. The batch size is set to 32. The length of each modality is 300, which is 16 after Linear compression. To solve the data imbalance problem (i.e., the observations of each category are unevenly distributed), we use cross-validation to eliminate potential interference caused by differences in driver. All data are randomly divided into 80% for training and 20% for testing.

4 Performance Evaluation

4.1 Baseline Models

To demonstrate the superior performance of Multi-TBP and its generalization across multiple tasks, we compared Multi-TBP with ML and DL methods. The details of the baseline models are described as follows:

Machine learning methods. We selected three machine learning models that are good at classification tasks for comparison. **KNN** [6]: This method uses proximity to classify or predict groups of individual data points. **RF** [3]: This approach is a supervised

Table 1: Performance comparison between machine learning, deep learning baseline model and Multi-TBP on EmoTake dataset. The best result is highlighted in bold, and the worst is highlighted underlined. Note: “W-F1 score” represents weighted F1-score.

Models		Quality		Reaction Time		Readiness	
		Accuracy	W-F1 score	Accuracy	W-F1 score	Accuracy	W-F1 score
Machine Learning	KNN	76.41	73.26	80.00	76.64	86.67	84.27
	RF	80.51	75.06	83.59	78.20	89.23	85.59
	SVM	82.56	74.68	83.08	75.40	90.77	86.38
Deep Learning	Linear-based	<u>72.82</u>	<u>71.59</u>	<u>75.38</u>	<u>74.46</u>	<u>84.62</u>	<u>83.67</u>
	LSTM-based	82.56	74.68	83.08	75.40	90.77	86.38
	iTransformer	84.10	77.54	86.67	85.15	90.77	87.04
	Multi-TBP (ours)	84.62	80.78	90.77	89.64	92.82	90.82

learning algorithm and an ensemble model composed of many decision trees. **SVM** [5]: This is a supervised learning binary linear classifier. The learning strategy is to maximize the interval in the feature space.

Deep learning methods. We select three deep learning frameworks designed commonly used for classification tasks. **Linear-based:** We use the Linear to build the framework. First, we use a Linear to unify the dimensions of each modality, then use multiple Linear and activation functions to predict takeover behavior. **LSTM-based** [19]: This method uses the LSTM and Linear to build the network. First, LSTM is used to capture temporal series information, and then the Linear are used for the prediction. **iTransformer** [23]: iTransformer applies the attention and feed-forward network to the inverted dimensions, and the attention mechanism is utilized to capture multivariate correlations.

4.2 Evaluation Metrics

We evaluated the performance of Multi-TBP on the data collected by EmoTake [18]. Since cross-validation can provide the overall performance of the classification task and show the performance of the model when the data partition changes in successive iterations, we used subject-independent cross-validation (5-fold) to eliminate the potential interference caused by different background information of the driver to evaluate the performance of predicting the driver’s takeover quality, reaction time, and readiness.

In existing studies, accuracy, weighted F1-score (W-F1 score), precision, and recall are often used to evaluate the system’s performance in classification tasks. However, since the weighted F1 score is more capable of evaluating unbalanced classification, we mainly use accuracy and weighted F1 score as the evaluation indicators of the model in subsequent experimental analysis. Their definitions are as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (2)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

$$\text{Weighted F1-score} = \sum 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \times W_i \quad (5)$$

where TP, TN, FP and FN denote True Positive, True Negative, False Positive and False Negative, respectively. Precision represents the

proportion of true positive examples in the prediction results. Recall denotes the proportion of all positive examples that are correctly predicted. W_i is the weight of the predicted category, and i is the number of categories.

4.3 Results and Analysis

We selected several ML and DL baselines for comparison in takeover behavior prediction, as shown in Table 1. The proposed method uses the Cross-modal Transformer for multimodal interaction and assigns different weights to each modality when fusion to determine their importance. Meanwhile, it adopts the multi-task learning strategy to capture the correlation information between tasks. Therefore, Multi-TBP achieves SOTA performance in all task metrics. This proves the necessity of bridging the gap between different modalities and mining the correlation between multiple tasks for takeover behavior prediction.

Takeover quality. As shown in Table 1, Multi-TBP outperforms all ML and DL models, especially the “W-F1 score”, which has significantly improved by 3.24-9.19. In addition, although the Linear-based method has the lowest performance, the overall performance distribution shows that the DL model is better than the ML model. This indicates that ML models often cannot effectively eliminate the gap between modalities due to the heterogeneity of multimodal data. Furthermore, among the three tasks, the performance of this task is much lower than that of the other two tasks, indicating that it is more difficult to predict the quality of the driver’s takeover. **Takeover reaction time.** For Reaction time, Multi-TBP has the most significant improvement, with an improvement of 4.1-15.39 and 4.49-15.12 in both indicators. **Takeover readiness.** Most models perform well in this task. However, the fact that SVM, LSTM-based, and iTransformer achieved 90.77% in “Accuracy” shows that the lower limit of predicting takeover readiness is very high, and it is difficult to improve the upper limit.

5 Discussion

5.1 Effects of Different Modules and Modalities

To explore the importance distribution of submodules and each modality, we designed the ablation study and added the methods (RF and iTransformer) used in existing studies [du, gu] for comparison, as shown in Table 2. for the importance of submodules (Effects of Each Module), when the multimodal fusion module (w/o

Table 2: Ablation results of Multi-TBP’s components (Effects of Each Module) and individual modality importance (Effects of Each Modality). The best result is highlighted in bold, and the worst is highlighted underlined. Note: “CrossTrans” is the Cross-modal Transformer; “Multi-task” represents the Multi-task learning strategy; “w/o” denotes “without”.

Methods		Quality		Reaction Time		Readiness	
		Accuracy	W-F1 score	Accuracy	W-F1 score	Accuracy	W-F1 score
Effects of Each Module							
w/o LSTM		83.08	75.66	86.15	82.03	<u>90.77</u>	<u>86.38</u>
w/o CrossTrans		<u>82.56</u>	<u>74.68</u>	<u>83.59</u>	<u>76.43</u>	91.03	86.75
w/o Multi-task		83.59	79.35	89.74	88.26	91.79	90.35
Multi-TBP (all)		84.62	80.78	90.77	89.64	92.82	90.82
Effects of Each Modality							
Action Units	RF	82.05	74.42	82.56	75.15	89.74	85.86
	iTransformer	81.03	73.91	81.54	75.19	90.77	86.38
	Multi-TBP	81.54	77.72	86.15	81.49	90.77	86.67
Body Posture	RF	79.49	73.11	82.56	78.45	<u>88.21</u>	<u>85.04</u>
	iTransformer	82.56	74.68	82.56	75.14	90.17	85.92
	Multi-TBP	81.02	76.48	83.58	83.57	88.72	86.66
Eye Movements	RF	78.97	73.38	78.46	<u>73.04</u>	90.77	88.42
	iTransformer	80.00	76.25	<u>75.90</u>	74.68	89.74	85.86
	Multi-TBP	80.76	76.49	84.61	83.50	90.26	86.12
Head Posture	RF	<u>76.92</u>	<u>72.32</u>	82.56	78.08	89.74	85.86
	iTransformer	80.19	73.37	81.03	76.13	90.59	86.13
	Multi-TBP	81.54	76.98	80.51	77.62	89.23	85.60

Table 3: Ablation results of task combination. The best result is highlighted in bold, and the worst is highlighted underlined. Note: ✓ represents whether the task is included in the current combination.

Task Combination			Quality		Reaction Time		Readiness	
Quality	Reaction Time	Readiness	Accuracy	W-F1 score	Accuracy	W-F1 score	Accuracy	W-F1 score
✓			<u>84.03</u>	80.58	—	—	—	—
	✓		—	—	<u>89.23</u>	<u>88.36</u>	—	—
		✓	—	—	—	—	<u>90.77</u>	<u>87.93</u>
✓	✓		84.62	79.96	90.26	89.38	—	—
✓		✓	84.62	<u>79.24</u>	—	—	91.28	90.28
	✓	✓	—	—	90.26	89.49	91.28	89.69
✓	✓	✓	84.62	80.78	90.77	89.64	92.82	90.82

CrossTrans) is deleted, the performance decreases significantly, indicating that the driver behaviors in different modalities express similar information. In addition, the performance also decreases after deleting the multi-task learning strategy (w/o Multi-task), indicating that there is a correlation between different tasks. In the importance of unimodal (Effects of Each Modality), we can see that the best and worst performance of each task are distributed in the four modalities, indicating that each modality provides different amounts of effective information for different tasks, among which facial expressions (Action Units) usually dominate, while body posture and eye movements sometimes also. Therefore, if the model only statically sets the importance of each modality and ignores the differences in different environments, the model performance will be further limited. Furthermore, the effect of each model in unimodal environment is significantly lower than that in a multimodal environment, indicating that compared with unimodal, the model in a multimodal scenario can extract takeover behavior clues

from different modalities. Notably, some indicators of RF have improved in unimodal environments, indicating that machine learning methods are more suitable for scenarios with relatively simple data. However, since autonomous driving is a complex scenario with a large amount of information, as shown in Table 1, the upper limit of RF is much lower than that of Multi-TBP and iTransformer.

5.2 Correlations between Multiple Tasks

Since most existing studies only predict a single task separately through multimodal data, and do not explore how to perform multiple tasks simultaneously to explore whether there is an impact between tasks, we performed ablation experiments on different task combinations, as shown in Table 3. It can be seen that the effect of multi-task has improved in all cases. This shows that when mining information of different tasks from the same data, tasks often interact with each other. There will be a decrease in the W-F1 score

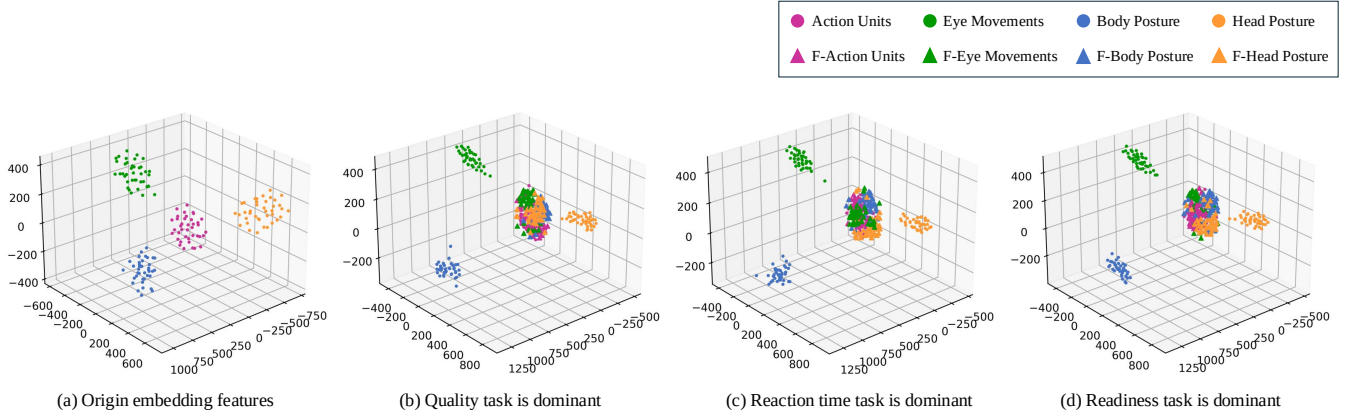


Figure 2: The t-SNE visualization [39] of the features distribution. The circle denotes the unimodal feature, and the triangle represents similar information that the unimodal feature provides in the multimodal feature.

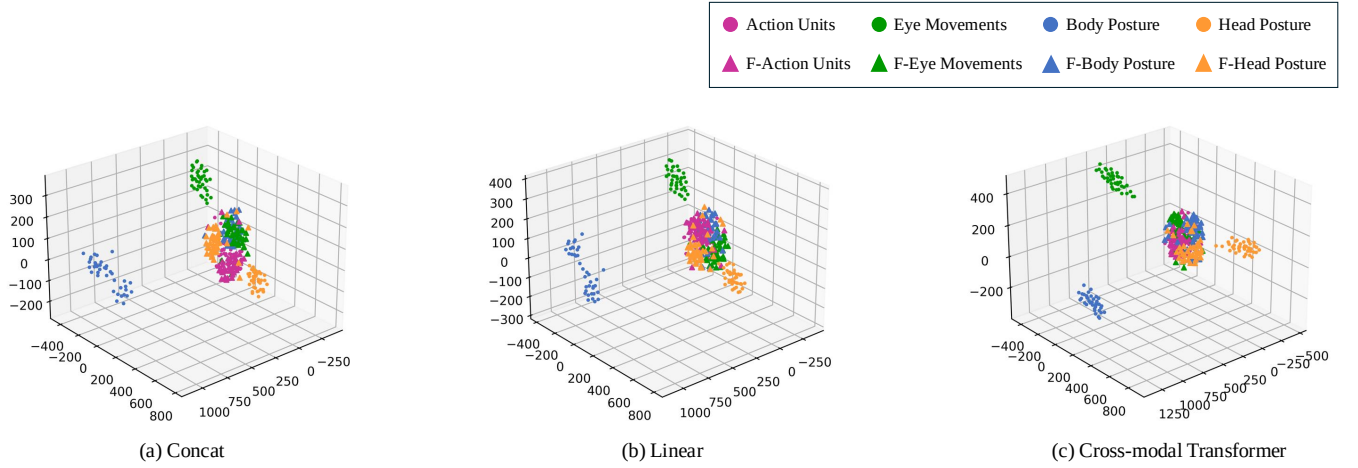


Figure 3: The t-SNE visualization of the distribution of the features. A different fusion method was used for each figure.

indicator of Quality, which may be because the correlation between Quality and other tasks is not always consistent. For example, when the takeover reaction time is short, the takeover quality may be high (simple scenario) or low.

5.3 Visualization of Features Distribution

To reflect the ability of Multi-TBP to bridge the gap between modalities, we visualized the unimodal features and the similar information of each modality in fusion features, as shown in Figure 2. From the distribution of the points in (a), it can be seen that there is a gap between each modality. From other figures, it can be seen that the multimodal fusion after Multi-TBP can effectively capture the similarity information of each modality (the triangles gather in the middle). In addition, it can be seen in (b), (c), and (d) that when each task is dominant, the similar information between modalities is closer to the Action Units (the triangle is close to the red circle),

which shows that the driver's facial expressions often provide more effective expression, thereby dominating the multimodal fusion.

5.4 Effects of Different Fusion Methods

To demonstrate the ability of Cross-modal Transformer to capture similar information between modalities, we selected methods commonly used in multimodal fusion in previous studies for comparison, as shown in Table 4. We can see that since Add directly adds all modalities, the information will be overridden, resulting in limited model performance. Concat and Linear treat all modalities equally, causing the model to be distracted by the secondary modality.

In addition, we also visualized the features obtained using different fusion methods, as shown in Figure 3. The regularity is similar to that in Table 4. Due to the limitations of the fusion method, the model cannot bridge the gap between modalities well (the triangles are not tightly clustered and are close to the circle).

Table 4: Ablation results of each fusion method. The best result is highlighted in bold, and the worst is highlighted underlined.

Methods	Quality		Reaction Time		Readiness	
	Accuracy	W-F1 score	Accuracy	W-F1 score	Accuracy	W-F1 score
Add	81.03	75.66	85.13	82.68	90.26	86.67
Concat	82.56	77.54	87.69	84.43	90.77	86.38
Linear	83.59	77.85	88.21	86.07	<u>90.26</u>	<u>86.12</u>
CrossTrans (ours)	84.62	80.78	90.77	89.64	92.82	90.82

6 Limitations and Future Work

Although the Multi-TBP has yielded exceptional outcomes and findings, some limitations remain. First, since autonomous driving technology requires accurate and real-time perception of the driver and the external environment to protect the vehicle's driving safety, the real-time computing efficiency and latency constraints are very strict [22]. However, ML and DL models consume computing resources and computing time during the training and inference stages, so deploying the model to the vehicle system and performing efficient real-time computing has become a challenge [44]. This study did not test the operating efficiency of the model in a real environment. To meet the challenges of real-world scenario diversity and real-time computing efficiency, we will consider using the system that fuses real and virtual driving scenarios in both AR and MR [29] and deploying the model in a real environment to perform real-time prediction, quantitatively analyzing its reasoning efficiency, and optimizing the model structure. Second, different TORs will also produce corresponding stimuli on the driver and affect the behavior, and the way TOR is released also needs to consider factors such as the driver, vehicle, and external environment. Therefore, in the future, we will design different TORs for the same driving situation to observe its takeover behavior and obtain more comprehensive driver data.

7 Conclusion

In this work, we proposed Multi-TBP to predict takeover behavior using multimodal data such as emotions collected from drivers. The proposed method uses Cross-modal Transofrmer to capture similar information between modalities and uses a multi-task learning strategy to mine the interactions when the same data performs different tasks. Next, to fully explore each modality's importance and multi-task learning's impact, we performed numerous ablation studies and visualization analyses to provide more information about how modalities and tasks affect takeover decisions. Our research shows that facial expression is critical for predicting takeover behavior. Meanwhile, there is a correlation between takeover quality, reaction time, and readiness.

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